Juan Esteban Agudelo Ortiz

Bogotá, Colombia | juan.es.agor@gmail.com | www.datascienceportfol.io/astroai_lab
linkedin.com/in/juanessao | github.com/juagudeloo

Summary

Data Scientist and ML Engineer with 4+ years of end-to-end experience building computer vision systems and large-scale data pipelines in production on GCP, AWS, and IBM Cloud. Proven track record deploying containerized model APIs (FastAPI, Docker) across multiple client-facing products, covering RGB, multispectral, and hyperspectral imagery across segmentation, classification, and regression tasks. Experience in LLM evaluation and RLHF task design through applied research work. Currently completing a PhD building a Visual Language Model for scientific image understanding, applying multimodal deep learning to large observational datasets. Recommended without reservation for senior ML roles by the COO of d_model (MIT). Active science communicator and data science content creator (@Datos.DeCiencia, YouTube).

Experience

Senior Data Scientist (Contractor), d_model – Remote — Feb 2026 – Mar 2026

- Designed adversarial evaluation protocols targeting 3 simultaneous LLM failure modes (sycophancy, bias, hallucination), extending red teaming coverage beyond standard single-axis assessments used in industry.
- Built prompt-based evaluation tasks that quantified comparative model performance across failure modes, enabling evidence-based model selection at the research stage.
- Recommended without reservation for senior ML roles by the COO upon contract conclusion (letter of recommendation available).

Data Scientist, Planet AI Space Group – Remote — July 2023 – Aug 2025

- Built and automated end-to-end computer vision pipelines processing 100+ GB of multi-temporal, multi-channel imagery (RGB, multispectral, and hyperspectral) at national scale across Colombia, reducing manual processing time by ∼90%.
- Designed and deployed large-scale ETL/ELT pipelines handling data ingestion, transformation, feature extraction, and structured output from satellite archives for downstream ML models; managed data workflows across multiple cloud environments.
- Deployed segmentation, classification, and regression models (CNNs, XGBoost) across 4+ production projects including vegetation assessment, water body detection (>95% accuracy), greenhouse gas concentration mapping, and Uber H3 multi-resolution land analysis, enabling new data-driven methodologies sold to external clients.
- Sole architect and developer of 3 production REST APIs (FastAPI, Docker, IBM Cloud) for model inference serving multiple client dashboards; established the team's cloud deployment standards for model APIs and interactive data products.
- Presented ML-based environmental analysis results directly to the Colombian Ministry of Sciences, securing six months of project funding through a competitive collaboration agreement.
- Co-authored a peer-reviewed article on national-scale mangrove carbon quantification using Random Forest on Sentinel-2 imagery (RACCEFYN, 2024; Scopus/WoS indexed); presented findings at RELACA and COP16.

Data Scientist, Fregata Space – Remote — July 2022 – July 2023

- Applied SRCNN-based image super-resolution to multi-channel satellite imagery targeting 4x spatial resolution enhancement (10m to 2.5m), evaluating output quality using PSNR and SSIM metrics across diverse geographic conditions.

Education

Universidad Nacional de Colombia, Ph.D. in Science in Astronomy — July 2025 – present

- Building a Visual Language Model (VLM) for scientific image understanding in solar physics, connecting a 366M parameter spatiotemporal transformer (NASA/IBM Surya) to an LLM via a learned projection layer.
- Developing a three-stage data pipeline combining LLM-based metadata extraction from 2,000+ scientific papers (Qwen2.5-14B-Instruct), astronomical archive querying, and image matching using normalized cross-correlation (NCC) with LightGlue refinement.

Universidad Nacional de Colombia, Master of Science in Astronomy — July 2023 – July 2025

- GPA: 4.9/5.0 (≈4.0/4.0 US scale); Meritorious Mention
- Thesis: Machine Learning Models for Inference in the Solar Atmosphere
- Advisor: Prof. Benjamin Calvo Mozo
- Designed and trained MUISCA, a physics-informed multi-scale CNN (PyTorch) on 700+ GB of solar atmosphere simulation data on an HPC cluster, recovering three atmospheric physical quantities across 21 depth levels; reduced inversion error by ∼38% compared to the classical baseline.

Universidad Nacional de Colombia, Bachelor in Physics — July 2017 – July 2023

- GPA: 4.5/5.0 (≈3.7/4.0 US scale)

Skills

Programming Languages: Python (proficient), SQL (experienced)

Technical Skills: Computer Vision, Machine Learning, Deep Learning (CNNs, PINNs), LLM Applications, Prompt Engineering, LLM Red Teaming, MLOps, Cloud Deployment, Data ETL/ELT Pipelines, Feature Engineering, Multimodal AI, Hyperparameter Tuning, Statistical Experimentation, Data Visualization, REST APIs

Technologies (production): PyTorch, scikit-learn, OpenCV, HuggingFace, Docker, FastAPI, GCP (Cloud Run, Cloud Storage, BigQuery), AWS (S3, EC2), IBM Cloud, Pandas, NumPy, TensorBoard, Streamlit, Plotly, Git

Technologies (projects/research): YOLO, LlamaIndex, LightGlue, Qwen, RAG

Professional Skills: Leadership, Technical Communication, Continuous Learning, Problem-Solving, Autonomous Work

Awards

Winner Astroco Award First Edition 2025 - Masters Thesis Category — Nov 2025

DynaSun Internship, Max Planck Institute for Solar System Research — 2025-02 - 2025-03

Academic Scholarship 50%, Universidad Nacional de Colombia — 2023-07 - 2023-12

Certifications

Natural Language Processing - Datacamp (2026): sentiment analysis, NER, recommenders for business applications

AI Agents Building - Datacamp (2025): LangChain, HuggingFace, LLM-based agents

Cloud Computing - Datacamp (2025): GCP, AWS, Azure basics for deploying ML models and apps

Big Data & Kubernetes - Datacamp (2025): ETL/ELT pipelines and MLOps fundamentals

PySpark for Big Data - Datacamp (2025): distributed data engineering and ML

HPC summer school - Universidad Nacional de Colombia (2024): Distributed computation for big-scale data storage and analysis

ESCAPE Summer School on Data Science for Astronomy, Astroparticle and Particle Physics - The European Science Cluster of Astronomy & Particle Physics (2021): Data science and Big Data knowledge for processing astrophysical and particle physics observations